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"Except where explicitly stated, all work in this report, is my own original work and has not been submitted elsewhere in fulfilment of the requirement of this or any other award"

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1 Introduction

This section will provide a brief overview on smartphones and social media. It will then provide context on the issue of smartphone addiction and social media addiction, including how people develop such addictions, and the effects they have on an individual. Existing apps and studies that attempt to deal with the issue are also mentioned. It will also provide an overview on the project and it's objectives to be achieved in the literature & technology review and during development of the app.

1.1 Project Background

Social media (SM) and smartphones have been adopted as an important piece of daily life for a substantial number of people in the world. They have provided us the ability to connect with from around the globe on a never-before-seen scale, allowing for friendships, communities, and relationships to be built from virtually anywhere. Dating back MySpace in the mid-2000s, the release of Apple's "iPhone" and the release of the HTC Dream running the "Android" operating system in 2008, to SM such as Twitter and Facebook reaching out to billions around the globe, smartphones and social media looks to be omnipresent within a majority of peoples' lives, with it being estimated that 1 in 3 people globally have some form of SM presence (Shah 2021), reportedly 85% of adults in the United States own a smartphone (Ouellette and Rao 2022).

Over the years, worries have been raised regarding many issues that concern smartphones and SM, particularly surrounding the problem of individuals developing an overdependence on their smartphones and SM that negatively affects their day-to-day life (Allen and Buzzi 2020; Primack et al. 2017). Commonly categorised as "Smartphone Addiction" (SA) and "Social Media Addiction" (SMA) and considered subsets of behavioural addiction (King and Dong 2017; Hou et al. 2019), the issue has been largely discussed and researched over the last 15 years, with researchers looking into potential causes and correlations with other health effects.

Indian researchers Bhalerao et al. (2020) analysed SA and provided potential causes behind SA, one of which being how the current generation has been brought up alongside the internet and technology, therefore possibly feeling a preference to communicate over the internet instead of in-person, fuelling the dwindling need for face-to-face communication. King and Dong (2017) made a series of hypotheses in relation to college students and the effects related to SA, looking at SA's associations with "life stressors" (a measure of stress and anxiety), "digital overload" (a measure of information and communication overload), retention, academic performance, and

social well-being. The results found that SA could be positively correlated with life stressors, digital overload, retention, and academic performance, something consistent with previous studies.

Potential treatments have been researched as well, A China-based study from 2019 analysing SMA (Hou et al. 2019) developed a cognitive-behavioural-therapy-based intervention program that used various techniques including reminder cards, journal entries before sleeping, and cognitive reconstruction. The results showed that compared to the control group, those who took part in the program saw improvements in mental health, engagement in learning, sleep quality, and also saw a drop in SM usage.

The "AntiSocial" app, developed by Australian company BugBean for Android devices allowed users to restrict access to certain apps and provided app usage statistics, but also added other functionality like usage comparisons with others that were within their age group, location, and occupation. It also showed users how many times they picked up their device, and how long they used certain apps for (Michael 2017). The author stated that the app ranked 3rd in downloads on the UK Google Play store mere days after launch, "demonstrat[ing] the need for [AntiSocial.io]" (ibid.).

However, there seemingly exists a gap of intervention apps that deal specifically with SA or SMA, and that also allow self-reflection from the user, and most literature on the topic mostly provide analysis on apps that deal with adjacent issues. A 2021 study by Oliveira et al. (2021) reviewed the effectiveness of psychological intervention mobile apps, and while they concluded that the apps showed promise, none of the literature specifically looked at treating the issue of SA or SMA.

1.2 Project Overview

1.2.1 Project Outline

Evidence has shown that smartphone apps that help with SA and SMA are in-demand, and significant amounts research have been conducted suggesting treatments to the issue. However, most consumer products on the app store opt to utilise app-restriction features. Because of this approach, young adults who use these apps aren't given much opportunity to reflect on the nature of SA & SMA and the potential causes of it, leading to very little character growth for the end-user. However, by taking a different approach by integrating psychology, young adults can be given the opportunity to reflect on their addiction and learn how to treat the

issue. With that being said, the research statement is as follows:

To create a mobile application that integrates psychological intervention methods alongside usage tracking technologies to help treat smartphone and social media addiction

1.2.2 Project Objectives

The primary aim of this project is to create an application that can guide the user into reducing their smartphone & social media usage with the use of usage tracking and psychological methods.

Secondary Objectives

Investigate existing mobile apps that use psychological methods to treat mental health issues
Research into how existing mobile applications make use of psychological methods when treating issues such as depression and anxiety to understand how viable using mobile apps are to treat similar issues.

Investigate the contextualisation of SA and SMA in existing apps
Research how the issue of SA and SMA are contextualised within existing intervention apps and what issues could arise from how they are presented. This will help in enabling growth in a user when it comes to learning about SA and SMA.

Investigate psychological frameworks
Research how psychological methods have been used in the past, specifically to treat SA or SMA, and see if such methods have been integrated with mobile apps to gain a further understanding on how these methods could be integrated or improved upon.

Investigate methods to track app usage
Research into how apps currently track phone activity, and how they can be utilised in the context of the project, as a major component of the application makes use of usage tracking.

Primary Objectives

Develop the smartphone intervention app
Create the application based on the knowledge gained from the literature and technology review. This will involve integrating the chosen psychological framework and usage tracking method before testing the app's functionality via test cases to ensure

the app functions as intended.

Gather participants for evaluating the app and perform evaluation

Participants will be used when evaluating the developed application. They will be given an overview of the app before being given a consent form to sign to consent to participating in the research. Participants will be split into two groups, one will be using the developed app while the other will be using an existing app that has app restriction functionality.

Analyse the results and obtain a conclusion
Collect results from the evaluation and see if the app was more impactful at reducing smartphone usage than traditional apps. Then draw up a conclusion based on the results, and see if the problem statement and hypothesis were met.

1.3 Project Hypothesis

Testing the hypothesis will involve splitting participants into two separate groups. One will attempt to reduce their smartphone & social media usage by using a mobile application with usage tracking and psychological intervention methods, while the other will do the same, but with a mobile application only utilising app restriction technologies. Given this, the hypothesis for this project is:

The combination of usage tracking and intervention methods rooted in psychology within a mobile application will help to reduce the smartphone & social media usage in young adults more than existing apps which only use app restriction technologies.

2 Literature and Technology Review

The literature and technology review is an integral component of the project as it allows the author to obtain foundational knowledge on the subject area that will then be used to carry out development of the primary research instrument, done by researching and analysing prior work surrounding the subject area.

This literature and technology review will review mobile apps that use psychological methods to treat mental health issues before looking at how existing apps provide context regarding SA & SMA and the issues associated with it. Potential psychological frameworks for treating SA & SMA are reviewed before researching how app usage can be tracked.

2.1 Investigate existing mobile apps that use psychological methods to treat mental health issues

The project background mentioned that although researchers have studied SA and SMA, few apps are available, or have been created for research purposes that handle the issue. However, apps that treat similar issues exist and have extensively been reviewed and evaluated. Therefore this section aims review existing studies on these apps to gain a further understanding on the viability of treating mental health issues.

2.1.1 Prior research on mobile app interventions

German researchers Weisel et al. (2019) reviewed mobile apps that integrated psychological frameworks to treat mental health issues such as depression, anxiety, drug use, PTSD, sleep-related issues, and suicidal ideation, with their intention being to evaluate the effectiveness of these apps in reducing mental health symptoms. The review discusses features used, with the most common being behaviour monitoring, participant engagement via explicit input, tailoring based on feedback, and gamification, while also noting the majority of apps opted to use CBT as their psychological framework. They discover that out of all the mental health issues targeted within the literature, depression, sleep-related issues, and smoking returned a significant effect, with the review stating that "our knowledge on how to design effective mental health apps is very much at the beginning".

They continue by suggesting different approaches that focus more on the user and the framework, as well as engagement with the app via usage patterns, before concluding by saying that at the current point in time, there isn't much significant evidence that supports using mobile apps for treating mental health as an alternative to traditional treatment, highlighting that while apps targeting depression,

sleep issues and smoking showed promising results, the results returned a high risk of bias within them (Weisel et al. 2019).

Portuguese researchers Oliveira et al. (2021) reviewed mobile app-based interventions utilising psychological frameworks targeting college students, aiming to review the outcomes of using app-based psychological interventions instead of traditional therapy, as well as looking how these apps are being embraced by students. The review details the types of mental health issues these apps target, while also noting that a majority of these apps are "self-guided" in nature (ibid.), with them looking to prevent the issue. Psychological frameworks used are mentioned, with cognitive-behavioural therapy (CBT) being the main framework used or combined with another framework, with other frameworks such as Acceptance and Commitment Therapy (ACT) and Self-determination Theory (SDT) mentioned.

The review then concludes by saying that the current literature demonstrates that using mobile apps for mental health intervention is a viable option and could be a great resource for counselling services, while noting that future work into the topic should utilise alternative psychological frameworks, as well as looking further into what students and therapists would want out of similar apps, and that future studies should stray from monetary incentives for students looking to participate in said studies (ibid.).

2.1.2 Discussion

Weisel et al. (2019) found that while apps that targeted depression, sleep-related issues, and smoking showed some results, the field was merely in its infancy when it came to designing effective applications that integrate psychological frameworks. Then, Oliveira et al. (2021) showed that mobile apps that targets mental health issues aimed towards college-level students were a viable alternative option for treating mental health issues due to student embracement, showcasing growth within the topic between the two points in time, though noted that this could've been due to incentives attached within the latter study.

Despite lacking insight on psychological frameworks other than CBT, showing a need to expand the types of psychological frameworks used for these apps, and the use of incentives which may bias user retention. the studies demonstrate that the viability of a mobile mental health app using psychological frameworks has grown, and that developing such app is doable, as long as aspects such as other psychological frameworks, and user incentive are considered.

2.2 Investigate the contextualisation of SA and SMA in existing apps

Near the end of the project background, it was mentioned that because of how intervention apps are developed with mostly functionality in mind, while neglecting any sort of teaching element, therefore leaving the user little to no room to learn about SA and SMA. This section intends to review how SA and SMA are currently contextualised, and how this could be improved.

2.2.1 Issues with contextualisation

Dosselaar (2017) puts forward an argument stating that SA apps created by developers aren't suitable for treating SA due to developers having limited knowledge on the topic of SA, and points out the irony in attempting to treat SA with the use of a smartphone. This sentiment is shared in Petrucco (2021), stating that proposing technology as a solution to an issue caused by technology itself is paradoxical in nature.

Dosselaar says that developers lack a deep understanding of the topic, and points out the overall complexity of the issue, mentioning how SA is a product of many other factors; not just the phone, and the apps simplify what is often a complex issue by acting as a one-size-fits-all solution (Dosselaar 2017; Roffarello and Russis 2019). They also fail to consider context behind smartphone usage, noting how it can be productive (Dosselaar 2017) and non-disruptive depending on where it's used (Petrucco 2021). Another issue is that the apps lack any kind of opportunity for the user to understand and reflect on the issue of SA, noting that these apps frame SA as an issue that can be quantified, while lacking any features that offer information about SA (Dosselaar 2017), something that plays a role in treating the issue (Rosa, Stracuzzi, and Faro 2022), and could influence app effectiveness (Rahmillah et al. 2023).

The author concludes by stating that SA apps are limited in effectiveness as they focus on practical functionality of the app rather than the contextualisation and opportunity for reflection for the user, before recommending researching integrating psychological frameworks into the apps (Dosselaar 2017).

2.2.2 Discussion

While it can be argued that some of these issues have been addressed, at a consumer level, most of these issues are yet to be addressed by anyone but the developers of these apps (ibid.). These apps focus only on features such as self-monitoring

and app restriction (Löchtefeld, Böhmer, and Ganev 2013), rather than providing opportunity for end-users to reflect and learn. These apps frame SA and SMA as an issue that can be solved with the apps themselves, rather than it being a complex issue (Santhosh 2025), emphasised by the fact that not all smartphone usage is the same for each person (Petrucco 2021), bringing suggestions of incorporating aspects such as psychological frameworks into these apps to make them more effective (Dosselaar 2017).

2.3 Investigate psychological frameworks

The project background mentioned that plenty of studies have been conducted utilising different psychological methods to treat SA and SMA. This section aims to review the most common psychological methods to understand which method would work best with the project.

2.3.1 Existing psychological frameworks for SA and SMA

The previous section highlighted the need for research into how psychological frameworks could be integrated into consumer SA and SMA intervention apps to make them more effective, a sentiment shared with Oliveira et al. (2021), who mentioned future intervention apps should incorporate [other] psychological methods.

Intervention apps developed for research purposes commonly use Cognitive Behavioural Therapy (CBT) as their psychological framework (Weisel et al. 2019; Oliveira et al. 2021), though other frameworks such as mindfulness, reality therapy, and theory of planned behaviour have also been utilised to mixed success (Gajecki et al. 2014; Witkiewitz et al. 2014; Lu et al. 2025).

2.3.2 Cognitive Behavioural Therapy

Cognitive Behavioural Therapy (CBT) is a psychological framework that take an "active, problem-focused, time-sensitive treatment approach" aiming to increase adaptive behaviour and reduce emotional distress in an individual with mental health issues via targeted, short-term strategies that intends to alter maladaptive emotional responses in an individual by changing their thoughts and/or behaviours (Wenzel 2017; Kaczkurkin and Foa 2015). Despite results lacking in some areas, it can be argued that it's the framework with the highest usage (Hofmann et al. 2012) overall, with some considering it the "gold standard" (Beck 2020) due to the proven success the framework has had on treating certain issues such as nicotine overdependence, schizophrenia, depression, and anxiety (Hofmann et al. 2012).

The main CBT techniques used in intervention apps are behavioural activation (An approach that focuses on engaging a patient in positive activities and helping with aversive control (Tindall et al. 2017)), cognitive restructuring (A process that assists patients in identifying and evaluating negative thoughts (Wenzel 2017)), and problem solving (Find and use solutions to a patient's issue (Denecke, Schmid, and Nüssli 2022)). Throuvala et al. (2020) assessed if the use of these techniques could reduce smartphone usage via three separate apps which provided these techniques (AntiSocial, Headspace, Pacifica) and found that those in the experimental group saw their smartphone usage drop and were made more aware of their usage when compared to the control group.

2.3.3 Reality Therapy

Reality Therapy (RT) is a psychological framework whose base principles can be traced back to choice theory, which itself theorises that all human behaviour is a result of choices, and life choices made are driven by our need for survival, love & belonging, power, freedom, and fun (Bradley 2014; Wubbolding 2016). RT's primary aim is on focusing a individual's behaviour in the present by using the Wants, Direction, Evaluation, and Planning & Commitment System (WDEP system) (Bradley 2014). This involves asking an individual to define a goal for themselves, what they're currently doing, and if that behaviour is halting progress towards their goal or not, before being encouraged to replace it with a healthier alternative (Yao et al. 2017). RT has shown effectiveness in treating issues such as Internet Addiction and Internet Gaming Disorder (Yao et al. 2017; Zajac et al. 2017), though has had limited research (Bradley 2014) when treating SA and SMA compared to CBT.

Rad and Ahmadi (2018) developed a mobile app integrating RT. The app was hypothesised to reduce anxiety, SM usage, among others, and was developed for Android devices. It implemented teachings about SMA, as well as app restriction functions, with the results being that anxiety and SM usage fell when compared to the control group.

2.4 Investigate methods to track app usage

The project background mentioned that existing SA and SMA intervention applications mainly make use of app-restriction and usage tracking technologies, and due to usage tracking being a large component of the project, it would be useful to know the technological methods used for app usage tracking.

2.4.1 Justification for app usage tracking

In some apps, app usage tracking has been implemented and shown to be successful in reducing smartphone usage, a study by researchers Rahmillah et al. (2023) review the effectiveness of apps intended to reduce Smartphone use, and found that in the 13 apps reviewed, 4 apps successfully reduced smartphone usage, those of which included app usage tracking, though it should be noted that all apps included some form of usage tracking.

2.4.2 Methods of app usage tracking

Rosa, Stracuzzi, and Faro (2022) hypothesised that individuals with high levels of smartphone usage show more difficulty in not using their smartphone, when compared to those with lower levels, verifying their hypothesis by developing an application called "SocialStatsApp" to track app usage of participants, alongside the use of questionnaires such as the Smartphone Addiction Scale-Short Version, used to screen adolescents who may be vulnerable to smartphone addiction (Kwon et al. 2013), as well as cognitive tests using external platforms. The app was developed for devices using the Android operating system, Google's mobile phone OS. The researchers mention how making the app compatible for iOS devices, Apple's smartphone OS, was not possible, as the OS disallows the collection of usage statistics for privacy reasons (Apple-Support n.d.). The programming language used was Java, one of three officially supported languages via the Android SDK, the other two being Kotlin and C++ (Android-Developers n.d.), though the study doesn't detail the reasoning behind using Java. For usage tracking, the app makes use of the Android class
android.app.usage.UsageStatsManager from the Android API which gives access to phone usage stats and history. Because of this, the app needs permission manually granted by the user. The results of the study showed that those with high levels of smartphone usage did experience more difficulty in not using their phone.

Researchers Bye et al. (2024) analysed the effects technology has on the mental health of young people, developing an app called the "3S-YP" app that would constantly collect what the researchers vaguely refer to as "smartphone metadata" from participants. While they don't detail the development of the app, they do mention that smartphone metadata was collected using the android.app.usage package from the Android API, while also mentioning that metadata collection was not performed for the iOS version of the application due to "data protection restrictions", a sentiment shared by the previous study, showing that developing for Android allowed for additional functionality not possible with iOS.

Outside of using the Android API, another method of tracking a user's app usage is by making use of screenshots and digital well-being features bundled with smartphones. Ubochi (2019) investigates the use of digital health apps and their effectiveness via Apple's Screen Time feature. Due to the aforementioned privacy restrictions implemented by apple, the researcher instructs participants to instead screenshot multiple aspects related to the Screen Time feature. A limitation of this method however is that screenshots must be taken at a specified time, as mistiming it could yield less accurate metrics (*Mobile sensing in psychology: Methods and applications* 2024).

2.4.3 Discussion

Both studies chose Android as their development platform of choice, mentioning that iOS apps don't allow for app usage tracking due to privacy restrictions (Rosa, Stracuzzi, and Faro 2022; Bye et al. 2024; Apple-Support n.d.). This is elaborated further in the book *Mobile sensing in psychology: Methods and applications* (2024) that mentions including iOS users would require a "jailbroken" phone, which brings a variety of issues with it. Meanwhile, Android allows for this, as long as the user permits it. Only one of the studies makes mention of a programming language (Rosa, Stracuzzi, and Faro 2022), that being Java, and while both studies make use of the android.app.usage package within the Android API, only one (ibid.) explicitly uses the UsageStatsManager class. Given this, it is clear that the OS that the app should be developed for is Android, and that using the android.app.usage package would be a viable method of app usage tracking.

3 Project's Primary Research Methods

This section will give a detailed overview on the methods that will be used to carry out the remainder of the project and provide a reasonable conclusion to the problem statement and hypothesis. This section will also outline any possible ethical issues, what work has been completed up to this point and all remaining work that will be carried out over the project's lifetime.

3.1 Research Method

This project aims to create a smartphone & social media addiction mobile intervention app that utilises psychological frameworks and usage tracking and find if it reduces smartphone & SM usage in young adults more than traditional intervention apps that use app-restriction features. The project will follow a develop and test methodology.

A mobile application that integrates a chosen psychological framework alongside usage tracking functionalities will be created, then at the conclusion of development, testing and evaluating the effectiveness of the app will begin, which will involve comparing the results of using the developed app for two weeks with an existing app that uses app-restriction functionalities for the same time period with participants. This testing method has been used by other researchers who have conducted research into the topic or have developed intervention apps (Throuvala et al. 2020; Rad and Ahmadi 2018). The results of this will then be evaluated to find if it satisfies the problem statement and hypothesis.

3.2 Develop the psychology-based intervention mobile application

3.2.1 Development

Due to the research carried out in the literature and technology review, the platform that the app will be developed for will be Google's Android OS. This is because Android applications easily allow for requesting usage statistics of apps installed on a user's phone via the android.app.usage package in the Android API, while Apple's iOS operating system doesn't allow for this due to privacy concerns. The app will be developed using the Java programming language, and will Android Studio as its IDE, as it's the IDE recommended by Google for developing Android apps.

Cognitive Behavioural Therapy and Reality Therapy were both discussed as psychological frameworks that could be used for the app, and from the research carried

out, Reality Therapy was decided on as the framework to integrate. This is because one of the main criticisms of SA & SMA intervention apps was the lack of opportunity to learn about SA & SMA, and Reality Therapy's aim is to make a patient aware on if their actions get in their way of their goals, and to then replace the action with a healthier alternative. Meanwhile, Cognitive Behavioural Therapy's methods don't innately provide teaching opportunities in its framework.

3.2.2 Project Lifecycle

A key component of any project is the lifecycle the project's development will follow. This project will make use of an agile methodology during the development. The Agile methodology is a framework that allows for iteration and flexibility throughout the software development lifecycle in response to user and stakeholder demands, an advantage over traditional methodologies such as the waterfall methodology (Al-Saqqa, Sawalha, and AbdelNabi 2020; Altameem 2015).

3.2.3 Testing

Testing will consist of integration testing and system testing. Integration testing will occur once the development of the prototype reaches certain development milestones, such as when implementation of a component is completed, where it will then be tested against a set of test cases. Development will then progress to the next milestone once the state of the component is at a satisfactory level. Near the end of development, the prototype will undergo system testing, where the entire prototype is tested from the perspective of an end-user for functional and non-functional elements (Loikkanen 2024).

3.3 Evaluation

Evaluating the application will involve gathering 10 participants to partake in testing the app. This is because the evaluation will involve splitting participants into two equal-sized groups, one of which will make use of the developed app, while the other will use a separate app that only provides app restriction in order to evaluate the app's effectiveness. After a two week testing period, participants will be asked to provide statistics about their app usage over the two weeks. Those in the group using the developed app will also receive a survey regarding their experience with the app. The results from this testing period will then be used to evaluate the app's effectiveness and identify any areas of further work.

3.4 Preliminary Work

Work surrounding the project is already underway. As of submission, requirements gathering is still underway, though is nearing the state of completion. Part of requirements gathering involved gathering responses via surveys and user interviews, though ethical approval has not been granted to conduct these at this time. Initial work on wireframes of screens to be included in the app is also underway.

3.5 Remaining Work

Below is a table containing a list of remaining tasks, their estimated start and end dates, a description of the task, and the expected deliverable at the completion of the task. It should be noted that check-ins with stakeholder(s) will occur at certain points throughout the entirety of the project.

Task Name	Description	Deliverable(s)	Start and End Date
Carry out requirements gathering	An extensive set of functional and non-functional requirements that satisfies the problem statement and hypothesis will be gathered and listed in a requirements specification document. This will consist of information found during the literature review, as well as independent research via one-to-one interviews and surveys.	Requirements specification document	Start: Ongoing as of October End: Est. Early November
Create design document	Once requirements gathering has concluded, the process of creating wireframes for the UI, flowcharts that deptict user choices and screen layouts will begin. These designs will then be used when developing the app.	Design document containing wireframes, flowcharts, screen layouts	Start: Est. Early November End: Est. Late November - Early December
Develop the application	Upon design approval from stakeholders, development of the prototype will begin, with the final result being an application that meets the requirements outlined in the requirements specification document.	Functional Prototype(s)	Start: Est. Early December End: Est. Mid February
Undergo testing for the application	Concurrent to development, the application will also undergo testing. When a component of the app is completed, integration testing will commence against a set of test cases to ensure the component works as expected. Integration testing will continue until all components are successfully implemented, where system testing will being, which tests the entire app from a user's perspective.	Results of testing inside a test document	Start: Est. Late December - Early January End: Est. Mid February
Conduct user evaluation	Evaluation will consist of splitting participants into two separate groups, one of which will use the app, while the other uses a separate app. After this, a survey will be given to those who used the developed app, assessing certain aspects of the app. Ethical approval is still pending as of writing.	Evaluation report document, final report	Start: Est. Late February End: Est. Mid-late April

Table 1: Remaining work to be carried out

3.6 Ethical Issues

The use of human participants for testing purposes was mentioned within the introduction when discussing the project's primary objectives. Participants will be used for this purpose and will play a significant role when assessing if the completed app meets the project hypothesis as they will be the ones testing the app. As well as this, participants are intended to be used for requirements gathering via user interviews and surveys. Due to this, ethical approval will need to be granted by the module co-ordinator before conducting anything involving participants. An ethical approval form was submitted, though as of the time of submission, approval has not been granted, nor has any response to the initial form has been given.

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